Sensors used in a Formula Student-Style Racing Car- Functionality, Integration and Indian Sourcing

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Abstract—This report surveys the sensors and onboard computer needed by a Formula Student-style racecar with an autonomous (driverless) stack. It explains how each device works (IMU and its accelerometer, gyroscope and magnetometer types; GNSS and the VectorNav VN-200 GNSS/INS; Hall-effect and optical wheel-speed sensors; mono and stereo cameras including the ZED 2i; the Hesai Pandar40P LiDAR; and the NVIDIA Jetson AGX Orin), what each contributes to our car, and what it costs. It then covers how the sensors are wired, time-synchronised and calibrated on the vehicle, how common faults are diagnosed, and where the parts can be bought from India, with prices quoted from vendor pages at the time of writing.

Keywords—Formula Student, autonomous vehicles, LiDAR, GNSS/INS, IMU, stereo vision, Hall-effect sensor, sensor fusion

I. INTRODUCTION

A driverless Formula Student car has to answer three questions every few milliseconds: where are the cones, where am I, and how fast and in which direction am I moving. No single sensor answers all three, so the car carries complementary sensors whose strengths and failure modes offset each other: LiDAR and cameras for the first, GNSS/INS for the second and wheel-speed, steering and brake sensors for the third. A central computer fuses the streams and sends torque, steering and brake requests to the actuators.

II. QUICK GLOSSARY

TABLE I
WORDS YOU WILL HEAR

Term	Plain Meaning
Sensor	A device that turns something real (motion, light, magnetism) into an electrical signal.
IMU	Inertial Measurement Unit: tells how the car is accelerating and turning.
GNSS/GPS	Satellite positioning. “GPS” is the American system; GNSS means all systems.
INS	IMU + GNSS + a smart filter, giving position, speed and direction
LiDAR	Laser distance scanner that builds a 3-D dot map.
Point Cloud	The cloud of 3-D dots that a LiDAR produces.
Stereo Camera	Two cameras side by side, like two eyes, to judge distance.
CAN bus	A two-wire network cars use so different units can talk.
ROS 2	Free robotics software framework we run on the computer.
Drift	A small error that keeps adding up over time.

III. IMU (INERTIAL MEASUREMENT UNIT)

What is it?

A tiny chip that feels motion. It has an accelerometer (feels speeding up, braking, bumps and gravity), a gyroscope (feels turning) and sometimes a magnetometer (a compass).

How does it work?

Inside an accelerometer is a microscopic weight on springs. When the car accelerates the weight moves slightly, and the chip measures that movement as a change in electrical capacitance. A MEMS gyroscope has a tiny vibrating structure; when it is rotated, the vibration is pushed sideways (the Coriolis effect) and the chip measures that push.

Accelerometer types: capacitive (cheap, accurate, used in almost every IMU), piezoresistive (good at huge shocks), piezoelectric (good at vibration, cannot feel steady gravity) and servo (very accurate but big and expensive). We use capacitive. Magnetometer types: Hall (cheap, weak), AMR (most common), GMR/TMR (more sensitive) and fluxgate (best, costly). On our car, magnets and high currents confuse the compass, so we do not trust it for heading.

Why do we need it?

It tells the car how it is tilting and turning, even where GPS is weak. It also helps correct LiDAR scans, which get blurred when the car moves while the LiDAR spins.

Catch: if you add up its readings to get position, small errors pile up (drift), so an IMU alone slowly goes wrong.

IV. GPS/GNSS

GPS is a satellite navigation system run by the USA. Your phone or car listens to signals from satellites, works out how far away each one is, and uses that to find where you are on Earth.

GNSS (Global Navigation Satellite System) is the general name for *all* such systems. GPS is one of them, like how "Xerox" is one brand of photocopier.

What: a receiver that listens to satellites. It measures how long each satellite's signal took to arrive and calculates where you are. **Accuracy:** normal GPS is good to a few metres, too coarse for a narrow track. GPS uses a fixed base station nearby to correct errors and reaches about centimetre accuracy.

Why we need it: to know our position, to record a "true path" to check our software against, and to provide accurate time. **Catch:** it does not work well next to tall walls or under roofs.

V. INS AND THE VECTORNAV VN-200

An **INS** combines the IMU (fast but drifts) with GPS (slow but never drifts) using a maths tool called a Kalman filter. The result is smooth, fast and correct position, speed and direction. The **VN-200** is a ready-made INS: it contains the IMU, a GNSS receiver and the filter in one small box. It is a very reliable choice. Prices: USD 2,700 (bare board) and USD 3,200 (metal housing) .

Easy tip: mount it firmly, near the car's centre, pointing the right way, and away from the motor and big cables. It has one GPS antenna, so it needs the car to be moving to work out its heading well.

VI. WHEEL-SPEED SENSORS, HALL EFFECT AND OPTICAL SENSORS

Hall effect. When electric current flows through a thin plate and a magnet comes close, a small voltage appears across the plate. That is the Hall effect. A Hall sensor sits near a toothed wheel (or a ring of magnets) that spins with the wheel. Each tooth passing makes a pulse. Count the pulses per second and you know the wheel speed:

speed = (pulses per second ÷ teeth) × wheel circumference (1)

Other types: variable-reluctance sensors are coils that need no power but do not work at very low speed; optical sensors shine light through slots in a disc and count the flashes. They are precise but get dirty easily. For a racecar, active Hall sensors are the simplest and most reliable. They cost roughly INR 500–3,000 each (estimate; check the shop).

Why we need it: it gives wheel and car speed, spots wheel slip, and double-checks what the INS says. Other optical sensors you may meet: steering encoders, infrared tyre-temperature sensors and expensive optical ground-speed sensors (too costly for us).

VII. CAMERAS

Mono camera: one lens, like one eye. Great for recognising colours and shapes (blue, yellow and orange cones). Hard to measure distance.

Stereo camera: two lenses a few centimetres apart. Close one eye and then the other: nearby objects "jump" more than far ones. The camera measures that jump to calculate distance. Distance gets less accurate the farther away the object is.

ZED 2i: a stereo camera with a wide 120° view, a waterproof (IP66) metal body and a built-in motion sensor. It plugs in via USB 3 and costs USD/EUR 499 from the maker (USD 739 with a polarizer from one reseller) . An Indian dealer listing for the older ZED 2 was INR 85,800 .

Why we need it: to tell cone colours and to see the track even when LiDAR returns are sparse.

Catch: it struggles in glare, darkness and heavy rain, and needs an NVIDIA GPU to run its software.

VIII. LIDAR (HESAI PANDAR40P)

What: LiDAR stands for Light Detection and Ranging. It flashes a laser, times how long the light takes to bounce back, and uses the speed of light to get distance. The Pandar40P has 40 lasers on a spinning head and gives a 360° ring of 3-D dots many times per second.

Why we need it: it measures distance to cones very accurately and works in any light. We group the dots into clusters and pick out the cone-sized ones.

Cost: it is an older model; used units are listed on eBay for roughly USD 179–550 .

Catch: it needs a stiff, vibration-free mount, can see rain as dots, and imports need customs clearance.

IX. THE COMPUTER: NVIDIA JETSON AGX ORIN

It is a small, powerful computer with a GPU for AI. It reads all sensors, runs the cone detector and the path planner, and sends commands to the car. The 64 GB developer kit has up to 275 TOPS of AI performance and runs Ubuntu 22.04 with JetPack 6. In India, ThinkRobotics lists it at about INR 2.08–2.09 lakh.

X. STEER-BY-WIRE AND BRAKE-BY-WIRE

“By-wire” means a motor, not a driver, moves the steering or the brake. To do that safely the car must know what the actuator is actually doing.

Steering: an angle sensor (magnetic or optical encoder) tells the steering position.

Brakes: a pressure sensor on the brake line and a position sensor on the actuator. Use two sensors where possible so they can check each other. If any of these stops working, the car must brake to a safe stop. An emergency brake system is required by the rules, so always read the current rulebook

XI. WHERE TO BUY

TABLE II

SOURCES AND PRICES (OCTOBER 2026)

Item	Where	Price seen
Jetson AGX Orin	ThinkRobotics (India) [11]	About INR 2.08–2.09 lakh
ZED 2i	Stereolabs store; IndiaMART dealers [5], [7]	USD 499; ZED 2 listing INR 85,800
VN-200	VectorNav store [3], [4]	USD 2,700 / 3,200
Pandar40P	eBay [8], [9]	USD 179–550 used
GNSS module	DigiKey/Mouser India [1], [2]	USD 108–277
Hall sensors, parts	Robu.in, Robocraze, Mouser/DigiKey India, Amazon.in	Estimate INR 500–3,000

Imported items attract customs duty and GST, often adding a third or more; confirm with the courier. Ask the college purchase office about concessions. Always get a GST invoice.

XII. CONCLUSION

Use LiDAR and cameras to see the cones, an INS for position and direction, wheel and actuator sensors for control and safety, and the Jetson to run it all. The costly items are the INS and the computer. Start by learning ROS 2, then play with a camera, then a LiDAR recording. Check the rulebook early.

XIII. ACKNOWLEDGEMENT

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